# Phase 2 - Week 5 - Lesson Day 1

ML Foundations

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## Welcome to Phase 2

Phase 1 taught you how to load, clean, explore, and test data. Phase 2 is where you build your first real **machine learning models** - the part most people think of when they hear "AI" or "data science."

## Today's Goal

- Understand what machine learning actually means
- Learn how to use Scikit-learn, the main ML library in Python
- Understand train/test splitting and why it matters

## 1. What Is Machine Learning?

**Simple definition:** Instead of writing exact rules by hand, you give the computer many examples (data), and it learns a pattern on its own. Then it uses that learned pattern to make predictions on new data it has never seen before.

Example: instead of writing rules like "if a house has 3 bedrooms, the price is $200,000," you give the model thousands of real house sales (size, bedrooms, location, price) and it learns the pattern between those features and price by itself.

**Supervised learning** is the type of machine learning we start with. "Supervised" means we give the model the correct answers during training (for example, the actual house prices), so it can learn from them. The two main types of supervised learning are **regression** (predicting a number, like a price) and **classification** (predicting a category, like yes/no or spam/not spam).

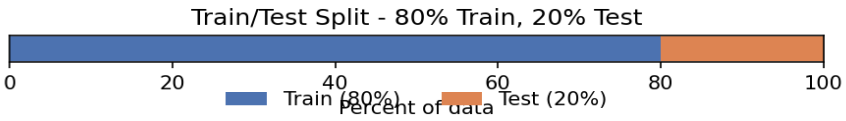
## 2. Scikit-learn - The Main ML Library

**Definition:** Scikit-learn (imported as **sklearn**) is the standard Python library for machine learning. It gives you ready-made models, so you don't need to build the math from scratch - you just prepare your data and call simple functions.

```
import sklearn
print(sklearn.__version__)   # confirms it's installed and shows the version
```

## 3. Train/Test Split

**Definition:** Before training a model, we split our data into two parts: a **training set** (used to teach the model) and a **test set** (kept hidden from the model, used only to check how well it learned).




**Why split the data at all?** If we test the model on the same data it trained on, it can look like it's doing great just by "memorizing" the answers - like a student who memorizes exam answers instead of understanding the subject. Testing on unseen data tells us if the model truly learned the pattern, or just memorized the examples.

```
from sklearn.model_selection import train_test_split

# X = the input features (what we use to predict)
# y = the target (what we are trying to predict)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42
)

# test_size=0.2 means 20% of the data goes to testing, 80% goes to training
# random_state=42 makes the split the same every time we run the code
```

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## Today's Practice

Open **day-1-ml-foundations.ipynb**. All answers are already solved with simple code and clear comments, using a small made-up dataset.

### Practice Questions (4 total - all solved)

**1. Confirm Scikit-learn works**

Import sklearn and print its version.

**2. Create a simple dataset**

Create a small made-up dataset with an input column and a target column (for example, hours studied -> exam score).

**3. Split the data**

Split your dataset into training (80%) and testing (20%) using train_test_split().

**4. Check the split sizes**

Print how many rows ended up in the training set and how many in the test set, and confirm they add up to the original total.

Once you and Halimo finish reviewing today's practice, move to Lesson Day 2 (Linear Regression) - your first real machine learning model.